FINAL REPORT

Factors Affecting Mental Health Outcomes

Data Analysis Graduation Project – 2025

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Course: Data Analyst – From Zero to Hero

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1. Introduction

Mental health has become a critical global public health issue, influenced by a complex interaction of demographic, clinical, lifestyle, and treatment-related factors. Understanding which factors are most strongly associated with mental health outcomes is essential for designing effective interventions and policies.

This project analyzes a global mental health dataset to explore how individual characteristics and contextual factors relate to overall mental health outcomes, measured on an ordinal scale from *Poor* to *Excellent*.

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2. Purpose and Objectives

2.1 Purpose

The purpose of this study is to apply data analysis techniques using Python to investigate factors influencing mental health outcomes at a global level. In order to examine how demographic, clinical, lifestyle, and treatment-related factors are associated with mental health outcomes

2.2 Objectives

- Analyze global mental health data to identify patterns and trends.
- Examine how demographic, clinical, lifestyle, and treatment-related factors are associated with mental health outcomes.
- Identify key factors that have the strongest impact on mental health.
- Provide actionable insights and recommendations to improve mental health care and well-being.

3. Data Description

3.1 Data Source

- **Dataset:** Global Mental Health Dataset 2025
- **Source:** Kaggle
- **Last Updated:** 18 December 2025

3.2 Dataset Overview

- **Number of records:** 2,500 (each record represents an anonymous patient)
- **Number of variables:** 15

3.3 Variables Description

- **Patient_ID:** Unique anonymized identifier assigned to each patient record
- **Age:** Age of the individual in years (18 to 80)
- **Gender:** Gender of the patient
- **Country:** Country of residence of the patient
- **Depression_Score:** Depression severity score based on the PHQ-9 scale (0–27)
- **Anxiety_Score:** Anxiety severity score based on the GAD-7 scale (0–21)
- **Stress_Level:** Self-reported stress level (Low, Medium, High, Severe)
- **Sleep_Hours :** Average number of hours slept per night
- **Physical_Activity:** Level of regular physical activity (None, Low, Moderate, High)
- **Chronic_Illness:** Indicates presence of any ongoing physical health condition (Yes/No)
- **Mental_Health_History:** Whether the patient has a prior mental health diagnosis (Yes/No)
- **Treatment:** Type of mental health treatment received (None, Therapy, Medication, Both)
- **Days_of_Treatment:** Total number of days the patient has undergone treatment
- **Outcome:** Overall treatment outcome or mental health status (Poor, Fair, Good, Excellent)
- **Work_Status:** Current employment or academic status of the patient

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4. Methodology

4.1 Tools and Technologies

- **Programming Language:** Python 3

- **Libraries:** pandas, matplotlib, seaborn, statsmodels, numpy, OrderedModel, scikit-learn

4.2 Data Cleaning and Preprocessing

The following steps were performed:

- Checked for duplicate patient identifiers to ensure each record represents a unique patient.
- Validated numerical ranges (e.g., age, depression and anxiety scores, sleep hours, day of treatment).
- Corrected data types for numerical and categorical variables.
- Handled missing values:
  - Filled missing *Physical_Activity* values using the mode.
  - Replaced missing *Treatment* values with **`"Unknown".`**
- Replaced missing *Treatment* values with `"Unknown"`

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5. Exploratory Data Analysis (EDA)

- To align with the research objectives, variables were grouped based on their real-world meaning:
  - **Demographic:** Who is affected (Age, Gender, Country, Work_Status)
  - **Clinical:** Health condition (Stress_Level, Depression_Score, Anxiety_Score, Chronic_Illness, Mental_Health_History)
  - **Lifestyle:** Daily habits (Sleep_Hours, Physical_Activity)
  - **Treatment:** Medical intervention (Treatment type, Days_of_Treatment)
- Visualization Techniques
  - Bar plots for categorical variables.
  - Box plots for numerical variables across outcome groups.
  - Percentage-based stacked bar charts to compare outcome distributions.
  - Heatmaps for correlation analysis.

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6. Key Findings from EDA

6.1 Demographic Factors

- **Country:** Mental health outcomes vary significantly by country.
  - Japan shows the highest proportion of *Excellent* outcomes and the lowest proportion of *Poor* outcomes.
  - South Africa and Brazil have relatively higher proportions of *Poor* outcomes.

- **Work_Status:**
    - Retired individuals show the highest proportion of *Good* outcomes.
    - Students have the highest proportion of *Poor* outcomes.
    - Employed individuals show a higher proportion of *Fair* outcomes.

6.2 Clinical Factors

- **Stress_Level** shows a clear negative relationship with mental health outcomes.
    - As stress increases from Low to Severe, the proportion of *Poor* and *Fair* outcomes increases.
    - The proportion of *Good* outcomes declines significantly in the Severe stress group.

6.3 Other Factors

- Depression score, anxiety score, lifestyle variables, and treatment-related variables show limited visible differences across outcome groups in EDA.

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7. Correlation Analysis

Spearman correlation was used due to the ordinal nature of the outcome variable.

Findings:

- Correlation coefficients between mental health outcomes and numerical variables are close to zero.
- This suggests no strong monotonic relationship between outcome and individual numeric factors such as age, sleep hours, or depression score.
- These results are consistent with overlapping boxplots observed during EDA.

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8. Ordinal Logistic Regression

8.1 Model Justification

Since the outcome variable is ordinal (Poor < Fair < Good < Excellent), ordinal logistic regression was selected instead of linear or multinomial regression.

8.2 Results Summary

- **Country (Japan)** is statistically significant:
    - Odds Ratio ≈ 1.56 → individuals in Japan have approximately **56% higher odds of achieving a better mental health outcome** compared to the reference country.
- Other demographic, clinical, lifestyle, and treatment variables were not statistically significant after controlling for all factors simultaneously.

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9. Hypotheses Based on Insights

- Mental health outcomes are shaped by broader environmental and social contexts rather than single behavioral or clinical indicators.
- Country-level factors and occupational stress may act as major confounding variables influencing mental health outcomes.

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10. Recommendations

- Results suggest that structural and contextual factors such as country and work status play a more important role in mental health outcomes than individual clinical or lifestyle variables. Mental health strategies should therefore prioritize country-level policies, workplace interventions, and preventive care rather than focusing solely on individual treatment.
- Employers and educational institutions should provide targeted mental health support, especially for ***students*** and ***employed*** individuals.
- Incorporating additional socioeconomic and contextual variables, such as income, access to healthcare, and social support, will help to better understand the determinants of mental health. Improvements in data collection and measurement should be made to more detailed data.

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11. Limitations

- Lack of socioeconomic and environmental variables limits explanatory power.
- Cross-sectional data does not allow causal inference.
- Mental health outcomes rely on self-reported measures.

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12. Conclusion

This study demonstrates that mental health outcomes are influenced more strongly by contextual factors than by individual clinical or lifestyle variables alone. Incorporating broader socioeconomic and environmental data is essential for a deeper understanding of global mental health.